

Self-Timed Physically Unclonable Functions

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Abstract—Physically unclonable functions (PUFs) exploit the physical characteristics of silicon and provide an alternative to storing digital encryption keys in non-volatile memory. In this seminal work, I propose for the first time the use of self-timed logic to implement PUFs. I describe two methods of using self-timed logic to implement novel PUF designs: SMARTPUF-I and SMARTPUF-II. The PUF designs I present address the challenges facing PUFs which prevent widespread adoption: robustness, entropy, light-weight architectures and configurability. And seek to enable next generation security-on-chip via “smart on-chip” structures.

Keywords—Physically Unclonable Functions; Asynchronous Physically Unclonable Functions; Self-Timed Physically Unclonable Functions; Asynchronous PUFs; Self-Timed PUFs.

I. INTRODUCTION

Many security mechanisms are based upon the concept of a secret. Classic cryptographic applications contain a secret key as input to encryption algorithms in order to scramble and decipher data.

While they are secure against attack at the algorithmic level, it is commonly known that digitally-stored secret keys can be attacked or cloned. In smartcards, keys are stored on-chip in non-volatile memory. Field-programmable gate arrays (FPGAs) instead store keys in off-chip memory, which requires extra-secure channels and protocols to transfer the secret keys.

Physically unclonable functions (PUFs) offer an efficient alternative to storing digital keys as a smart on-chip structure by exploiting physical manufacturing variation of chips; and can enable a new generation of security-on-chip. A PUF maps a unique set of digital inputs, known as *challenges*, to a corresponding set of digital outputs, known as *responses*, for use in challenge-response security protocols.

Almost every year since 2000 there has been a new PUF design proposed as highlighted in Table I, while a typical identification authentication scenario is as illustrated in Figure 1. Here, a challenge is given to a chip to authenticate its identity. If the received response is not equal to the known challenge (recorded during manufacturing) it is identified as fake.

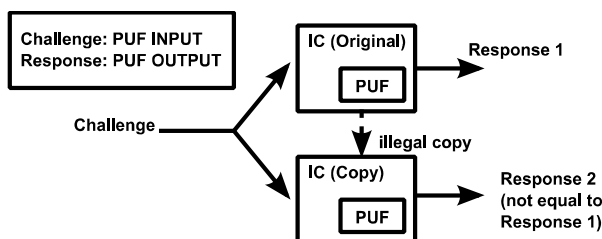


Figure 1. Verifying chip identity using a PUF

The unique benefits of PUFs come with inherent stability design issues. In addition, in their basic configuration PUFs lack enough entropy to prevent modelling attacks [1]. However, it can be observed that PUFs are naturally asynchronous in nature, insomuch that they attempt to exploit non-synchronous effects such as meta-stability, propagation delay or data glitches. This has led us to believe and hypothesise that self-timed techniques, widely known for robustness and high-entropy (e.g. random number generation), can deliver much better PUF designs. In this paper, I propose for the first time, using self-timed logic to implement PUFs to improve the existing state-of-the-art, and to enable next generation security-on-chip via “smart on-chip” structures.

To validate this hypothesis, I describe two methods of using self-timed logic, as “smart on-chip” structures, to implement novel PUF designs. SMARTPUF-I allows increased robustness and delivers increased entropy from its inherent configurability. It uses self-timed rings (STRs) to replace inverter ring oscillators (IROs) used in ring oscillator PUFs [2] (RO-PUFs).

While SMARTPUF-II completely removes the IROs used in RO-PUFs. It exploits the fact that the counters normally clocked by IROs in RO-PUFs can be made self-timed. Hence, this particular design is extremely light-weight with respect to silicon metrics - indeed the lightest to-date the authors believe. Furthermore, it may remove many of the issues associated with use of IROs in PUFs e.g. injection locking effects.

Our research, technical and scientific contributions in this seminal work include, but are not limited to, as follows:

- The use of self-timed logic for a new class of PUFs is proposed.
- New PUF designs are proposed as “smart on-chip” structures to enable next generation security-on-chip.

The remainder of the paper is organised as follows: Section II gives an overview of self-timed logic. Section III describes SMARTPUF-I. Section IV describes SMARTPUF-II. Section V draws conclusions.

Year	PUF Type
2000-2004	Device mismatch [8], One-way function [9], Physical Random Function [10], Arbiter PUF[11]
2005-2008	Coating PUF[12], Ring Oscillator PUF[2], SRAM PUF[13], Butterfly PUF[14]
2009-2011	Power distribution PUF[15], Glitch PUF[16], Mecca PUF[17]
2012	Self-timed / Asynchronous PUFs - SMARTPUFs (this paper)

TABLE I. DIFFERENT TYPES OF PUF

- $N_b \geq 1$, where N_b is the number of bubbles
- N_t is a positive even number of tokens

3) Noise

Both STRs and IROs will exhibit Gaussian jitter such that the propagation delay will resemble a normal distribution. However, where they differ is through how jitter accumulates. An IRO's period is defined by two loops of one token around the ring and accumulates jitter from the number of crossed stages; but in a STR, several tokens propagate in the ring simultaneously indicating the period is governed by the time between successive tokens. As such, each token crossing a stage experiences a variation in its propagation delay due to the Gaussian jitter contribution of that particular stage. This is contrary to the IRO effect of jitter accumulation.

In addition to Gaussian jitter, deterministic jitter occurs from non-random variations in propagation delays due to external global influences. The main difference again is that in a STR, several events propagate simultaneously, while deterministic jitter affects each event in the same way rather than affecting the whole structure. This leads to increased robustness in STRs versus IROs and a specific benefit of SMARTPUF-I as it is more immune to jitter.

B. Design and analysis of SMARTPUF-I

In this sub-section, I present a way to build a self-timed PUF, known as SMARTPUF-I, using self-timed logic based on self-timed rings by replacing IROs in RO-PUFs.

RO-PUFs are composed of two identically laid-out ROs, RO_1 to RO_2 , with frequencies f_1 to f_2 . They are selected using a pair of multiplexers that takes a bit of the PUF challenge as the select bit. Due to process variation, f_1 and f_2 will differ, generating a response bit R from comparison of the two frequencies measured by their respective counters.

When enabled, R will be 1 if $f_1 > f_2$, otherwise 0, hence producing a single bit of a PUF signature. The exemplary design in Figure 6 produces a single PUF bit - n -bit PUF configurations are built by cascading these basic structures.

Figure 8 illustrates the design of SMARTPUF-I which is differently composed of two STRs, but which is similarly selected by the PUF challenge via two multiplexers. Whichever STR is the fastest will determine the value of R .

For correct operation, the STR run-time has to be low enough so that the counters do not overflow. Hence, care has to be taken to ensure the counters are matched to the estimated frequencies.

The mechanism to enable or disable the pairs of IROs also needs to be correctly synchronised. It is worth noting that other methods are perfectly plausible to convert the varying STR frequencies to a binary bit, rather than using a pure multiplexer approach. How RO-PUFs are cascaded for n -bit PUFs may also differ (e.g. STR re-use).

Since IRO frequencies are closely matched, environmental effects can cause the oscillators to switch their outputs, with increasing temperature and/or decreasing voltage resulting in incorrect responses. It can also be observed that large arrays of ring oscillators can cause a change in local chip temperature.

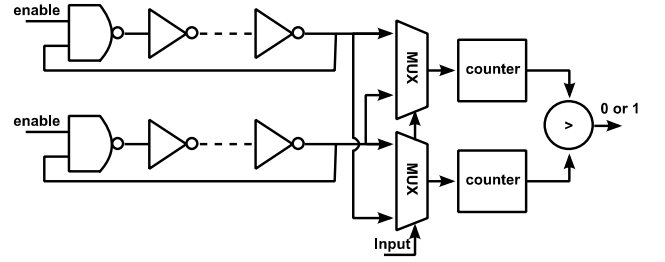
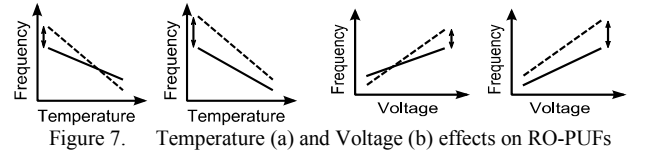


Figure 6. Ring-oscillator based PUF design

These stability issues are depicted on the left in Figure 7(a) and (b). Ideally, the frequency difference should be sufficient to ensure consistent operation over temperature and voltage shown on the right in Figure 7(a) and (b).

The solution to this problem in normal PUFs is to use error-correcting methods, which are expensive in terms of silicon metrics and add additional complexity to the challenge-response protocol [1]. This can be avoided in a SMARTPUF-I through the way STR pairs are configured via N_t and N_b ; that is, by purposely controlling L , the number of stages, and their initial value by setting or resetting the Muller-C elements through a load input as shown in Figure 8. For instance, we could configure a SMARTPUF-I with $L = 5$, and load the bit pattern {10111}, which is analogous in terms of tokens (T) and bubbles (B) to {TTBBB}. Note that all the STRs have to be loaded with the same initial value. The only constraints are that for oscillatory behaviour, the rules for oscillation must be obeyed.



This can be exploited during manufacturing for increased robustness in addition to the natural robustness to jitter noise. By determining the configuration of the STRs with the maximum frequencies, maximum reliability can be attained. This inherent configurable permits extremely low error rates by tending towards the ideal scenario. A further opportunity is to calibrate the SMARTPUF-I's configurability according to different operating conditions. For example, the entire operating range of temperature and voltage could be divided into regions and have different STR load bit patterns.

The other disadvantage of RO-PUFs is that they can be easily modelled to break the underlying security [1]. Usually this is through modelling a manufactured silicon batch for later cloning. However, SMARTPUF-I offers the opportunity to not only increase robustness through tolerance to environmental effects, but also the fact that it can be re-configured (i.e. it is naturally field-programmable) increases entropy to address modelling attacks. Since, as discussed, STRs can be easily configured to change their frequency by controlling N_t and N_b .

Thus, varying N_t and the load bit patterns in-field will result in completely new PUF designs, therefore thwarting modelling, as no two PUFs are the same. Another alternative

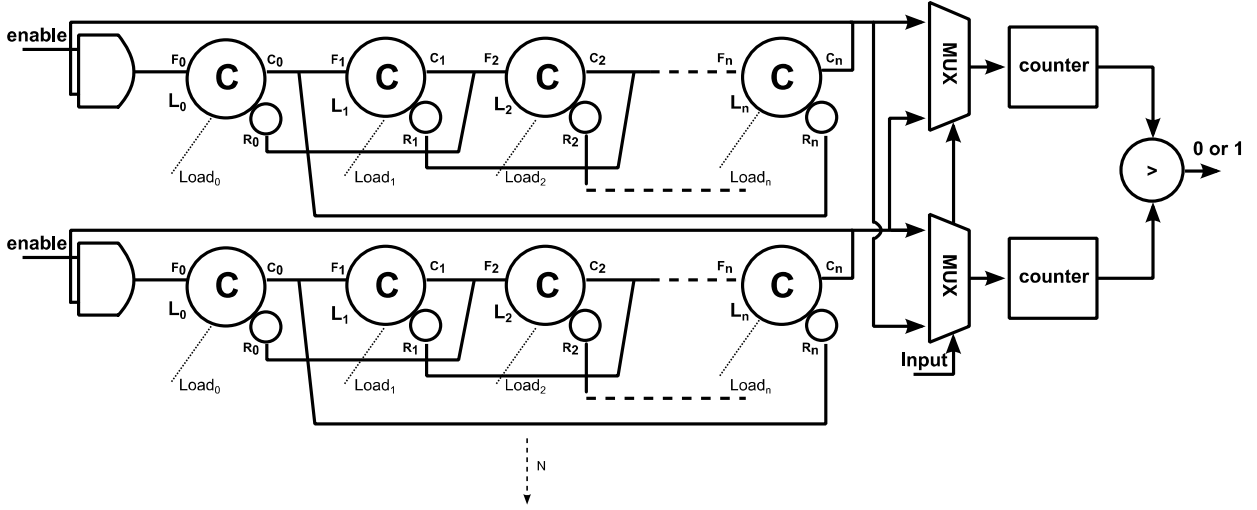


Figure 8. SMARTPUF-I design

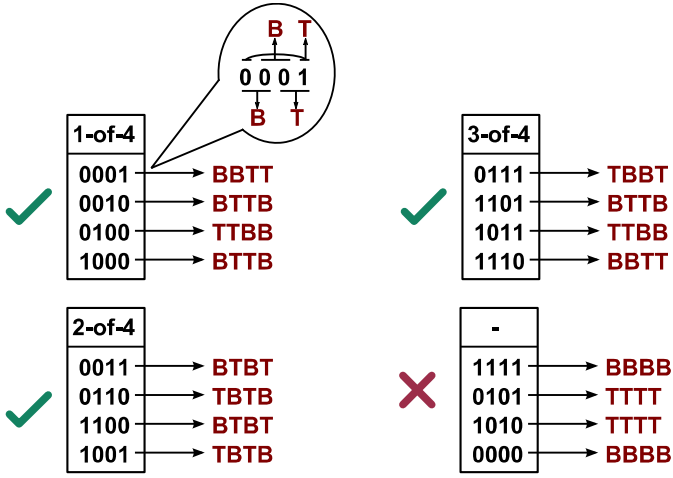


Figure 9. Four stage STR configurations

is to allocate different bit patterns randomly during manufacture and store in on-chip non-volatile memory.

Given STRs are not like other field-programmable designs and certain conditions are necessary to ensure oscillatory behaviour. It is worth demonstrating visually how the initial binary load patterns map to tokens and bubbles. Figure 9 illustrates the possible states for a four stage STR and expresses the possible codes in an m-of-n format – which is a common data representation in self-timed design.

IV. SMARTPUF-II

In this section, I present another smart on-chip PUF design that can also enable next generation security-on-chip.

We have observed in experiments that the counters present in normal RO-PUFs are subject to manufacturing variability just as much as IROs. This stands to reason given they are both composed of MOS transistors. Indeed, PUF counters are potentially larger structures versus their IROs. Hence, it is further plausible they may bear stronger components of variability, or moreover amplify the effects of variability versus IROs. This has led us to postulate that if the counters could be made to operate on-demand, that is, start counting on the

activation of an enable signal they could be directly harnessed. Thus, one can see, quite clearly, how this would deliver:

- from an inventive point of view, a very innovative and novel PUF design;
- and a silicon metric stand-point, a highly compact and light-weight design.

This is, explicitly, because the use for IROs can be completely removed thereby minimising the silicon area.

I have taken this approach to implement SmartPUF-II, which uses two self-clocking counters based on self-timed delay-insensitive logic to form a 1-bit PUF response. It can be appreciated that there is more than one way to implement such a counter. Therefore, the following implementation may not be the only solution, since the self-timed discipline allows for a plethora of options. For instance, a dynamic logic version could be used instead.

The implementation of a self-clocking counter [18] is shown in Figure 10. It consists of n toggle flip-flops, $n-1$ inverters and a AND gate to form an n -bit counter; where each block corresponds to a data bit. The gate structure of one implementation of a toggle flip-flop, as described and proposed in [18], is shown in Figure 11. Note, we are not proposing self-clocking counters or self-timed toggle flip-flops. Various solutions are available in literature as mentioned in the previous paragraph. Our contribution and improvement on the existing state-of-the-art is in their application to implement a very light-weight PUF design, namely SMARTPUF-II, with respect to silicon metrics.

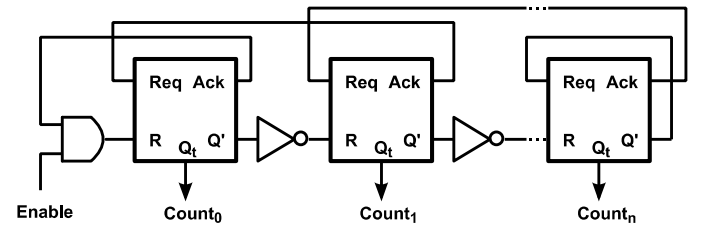


Figure 10. Self-clocking counter

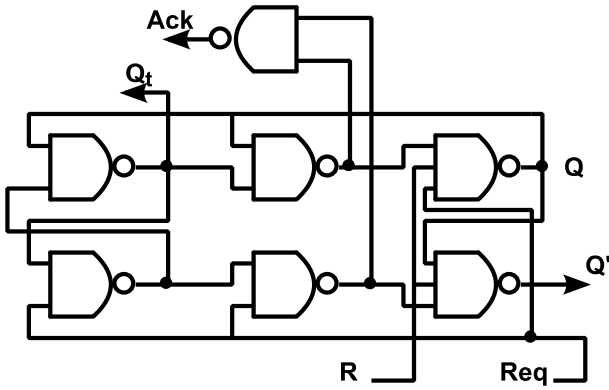


Figure 11. Self-timed toggle flip-flop

By using two of these counters, two multiplexers and a comparator 1-bit of SMARTPUF-II can be implemented as shown in Figure 12. An n -bit SMARTPUF-II is implemented by cascading the 1-bit design in Figure 12.

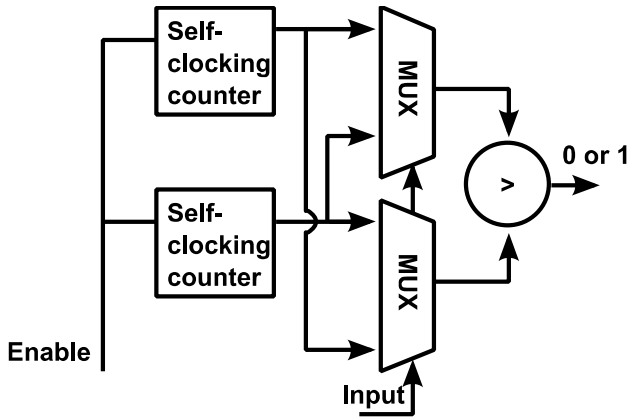


Figure 12. SMARTPUF-II design

V. CONCLUSION

I have proposed using self-timed logic to address the inherent issues with physically unclonable functions in this seminal work. Two new novel PUF designs have been presented and described. SMARTPUF-I is a design that uses self-timed rings to enable increased robustness and entropy. While SmartPUF-II is a design that delivers a very light-weight PUF architecture. As cryptographic primitives, PUFs have several useful applications in security but are most frequently used for device authentication (i.e. Figure 1). However, a new level of robustness, entropy and size allows increased resistance to modelling attacks and makes it feasible for real-life efficient design. And also enables new applications: secret key generation, Intellectual Property (IP) protection, and to prevent product counterfeiting – or even to use in a software and hardware scenario.

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